

Kabeen Kim

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Research Interests: Knowledge Graphs · GraphRAG · Multi-hop Retrieval · Evidence Reranking · Grounded LLM Reasoning

EDUCATION

DUKSUNG WOMEN'S UNIVERSITY

B.S. in Computer Engineering & Information Statistics (4.15/4.5)

Seoul, S.Korea

Mar 2021 – Feb 2027

UNIVERSITY OF MISSISSIPPI

Exchange Student, Computer Science

Mississippi, USA

Aug 2024 – Dec 2024

PUBLICATIONS

- Haopeng Chen, Yihao Ai, **Kabeen Kim**, et al. “UDAPose: Unsupervised Domain Adaptation for Low-Light Human Pose Estimation.” *Computer Vision and Pattern Recognition (CVPR)*, 2026.
- **Kabeen Kim***, Yena Kim*, Minjeong Seo*. “ConFuse: Context-Aware Fusion of LLMs and GBDTs for Sleep Prediction from Lifelog Data.” *International Conference on ICT Convergence (ICTC)*, 2026. *Equal contribution.
- **Kabeen Kim**, et al. “SPTC (Single-Pass Tree Chain-of-Thought): An Efficient Prompting Scheme for Small Language Models.” *Korea Computer Congress (KCC)*, 2025.
- **Kabeen Kim**, et al. “CoDU: Multi-Stage Approach for Complex Document Structure Understanding.” *Korea Data Base Conference (KDBC)*, 2025.

RESEARCH EXPERIENCE

ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE (ETRI)

Daejeon, South Korea

Research Intern - Knowledge Graph & Multi-Hop GraphRAG

Jul 2026 – Aug 2026

- **Raised the share of questions retaining all gold evidence from 40.0% to 76.8% (100/250 → 192/250; 1.92x)**. Identified stateless graph expansion and independent chunk-level reranking as the primary causes of multi-hop evidence loss, then redesigned both stages.
- Implemented a **top-40 path-state-preserving beam search** with cumulative cross-hop scoring (**0.75 prior path + 0.25 current step**). Built a path-aware reranking module using **BGE-reranker-v2-m3** to score question–path pairs and reconstruct Top-7 context from evidence recurring across high-scoring paths, eliminating gold-evidence loss during reranking.
- Developed a grounded generation pipeline using **Qwen2.5-32B-Instruct** with Top-5 reasoning paths and Top-7 evidence chunks. Improved LLM-judge **correctness 2.84 → 3.86, completeness from 2.88 to 3.91, and provenance grounding 2.83 → 3.89**. Outperformed KG²RAG, HippoRAG, and Microsoft GraphRAG across the evaluated judge dimensions on the in-domain doctrine benchmark.
- Built an evidence-verified OpenIE knowledge graph from U.S. Army doctrine using NetworkX and a Qdrant vector index, yielding **~366K total edges and ~38K semantic edges**. **Curated a 250-question multi-hop QA benchmark** comprising 150 two-hop and 100 three-hop questions.

DMKD LAB, DUKSUNG WOMEN'S UNIVERSITY

Seoul, South Korea

Undergraduate Researcher - LLM Reasoning, Document Understanding

Mar 2025 – Feb 2026

- Designed **SPTC**, a generation → self-evaluation → selection scheme for efficient reasoning in small language models, achieving **~1.5× faster inference** than Tree-of-Thought on CommonsenseQA/OpenBookQA; published as first author at **KCC 2025**.
- Built a document-structuring pipeline (DocLayout-YOLO fine-tuning, bounding-box refinement, reading-order reconstruction) that **improved layout-detection F1 by +0.475, placing 3rd / 210** at the Samsung AI Challenge and leading to first-author work CoDU (**KDBC 2025**).

COMPUTER VISION LAB, UNIVERSITY OF MISSISSIPPI

Mississippi, USA

Undergraduate Researcher - Low-Light Human Pose Estimation

Aug 2024 – Feb 2025

- **Contributed to UDAPose (CVPR 2026)** through domain-adaptation fine-tuning, training-pipeline tuning, and benchmarking against CycleGAN, UNIT, ELLA, and UNSB.
- Developed a **15-category failure taxonomy** to identify model and dataset weaknesses and guide subsequent model analysis and data-collection strategy.

INDUSTRY EXPERIENCE

BOSTON CONSULTING GROUP (BCG)

Seoul, South Korea

Research Analyst Intern

Mar 2026 – May 2026

- Built analysis-ready datasets from **30M+ rows of battery-manufacturing process data** using SQL, Python, AWS S3, and Spotfire. Trained XGBoost and LightGBM models that improved predictive performance by **~10%**.
- Fine-tuned a domain-specific **Qwen3-14B** model and benchmarked 10 LLMs, raising LLM-as-a-judge scores **2.3 → 3.3**. Automated AI-assisted research and reporting workflows using Claude Enterprise.

PROJECTS

RAG-BASED AI AGENT FOR PRODUCT RECOMMENDATION

May 2025 - Aug 2025

- Built an end-to-end **RAG product-retrieval pipeline** that crawled Samsung’s official website, normalized heterogeneous product specifications, and indexed product knowledge in Pinecone.
- Designed chunking and history-aware query rewriting and integrated a Cohere reranker. Improved retrieval **precision from 0.40 to 0.70** through reranking, query normalization, and few-shot prompting.

IMMERSIVE MULTISENSORY EXHIBITION SYSTEM USING MULTIMODAL AI

Feb 2025 - Nov 2025

- Built an end-to-end **multimodal AI system** combining CNN-based visual emotion recognition, CLIP semantic understanding, and MusicGen to transform an input image into synchronized audio and haptic experiences.
- Designed **YOLO-based input routing** and asynchronous inference, achieving **30 s end-to-end latency**. Converted RMS, bass, and onset features into synchronized vibration patterns.

ACTIVITIES

TOBIG’S 25TH COHORT - AI MEMBER

Jan 2026 - Dec 2026

- Collaborated with AI students from multiple universities to study **LLM architectures and prompt engineering**, and applied them through a **Codex Hackathon** and a **multi-agent reinforcement learning project**.

DUKSUNG GLOBAL CHALLENGER - TEAM LEAD, “BE A CONNECTOR”

Mar 2025 - Jul 2025

- Led a **four-member team, KRW 10 million university-funded global project**, overseeing the end-to-end planning and prototyping of a beacon-based campus–local business engagement app.
- Conducted field research and stakeholder interviews across the **UK, France, Germany, and the Netherlands** for service validation and benchmarking.

CJ UNIT, 8TH STUDENT VOLUNTEER - VOLUNTEER CODING INSTRUCTOR

Mar 2023 - Jun 2023

- Developed SW curricula and delivered remote coding instruction, including LEGO-based programming activities, to local middle-school students.

HONORS & AWARDS

- 3rd Place, Samsung AI Challenge - Visually-Rich Document Understanding (3rd of 210 teams), 2025
- 31st Mirae Asset Overseas Exchange Scholarship, Mirae Asset Park Hyeon Joo Foundation, 2024
- Excellence Award, TAVE 13th Data Analysis Project, 2024
- 2nd Prize, DS Global Vision Makers Innovation Idea Competition, 2024
- Academic Excellence Scholarship, Duksung Women’s University, 2021–2022 (3 awards)

TECHNICAL SKILLS

- Knowledge Graph & Retrieval:** Knowledge Graphs, GraphRAG, multi-hop retrieval, graph traversal, evidence reranking, hybrid graph–vector retrieval; NetworkX, Qdrant, BGE-M3
- LLM & RAG:** Qwen2.5/Qwen3 fine-tuning, QLoRA, RAG, prompt engineering, LLM-as-a-judge evaluation; LangChain
- Machine Learning & Computer Vision:** PyTorch, XGBoost, LightGBM, DocLayout-YOLO, OpenCV, CLIP, YOLO
- Programming & Systems:** Python, C++, SQL; AWS EC2/S3, Docker, Git, Weights & Biases, n8n

CERTIFICATIONS & LANGUAGES

- SQLD (SQL Developer)
- IELTS 7.0
- OPIc AL